

HDI Predictor Documentation

Coding & Solution

Date	03 July 2025
Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
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This section evaluates the quality and completeness of the implemented code and the overall solution.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/saivenkata-A7/hdi-predictor <i>(placeholder — update with the actual repository URL)</i>
Programming Language(s)	Python
Framework(s) Used	Flask, scikit-learn, Pandas, NumPy, Docker
Key Features Implemented	HDI tier classification (Very High/High/Medium/Low), HDI score estimate, feature-importance summary, Flask web form, Dockerized deployment
Pending / Incomplete Features	Multi-country side-by-side comparison view (stretch goal, not required for core deliverable)
Setup / Run Instructions	See “Project Executable Files” document for full step-by-step setup

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations (form input, model loading)	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The model was trained using `RandomForestClassifier` with class balancing to handle the natural skew toward “High” and “Very High” categories in the global country dataset, achieving **97.6% accuracy** on the held-out test set. The final model and scaler are serialized with `joblib` and loaded once at Flask startup to keep inference latency low.